

Decent Coaching Classes

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Tuesday Holiday Homework

Part I: Ch 1-4 (Linear Equations, Quadratic Equations, AP, Financial Planning) | Part II: Ch 1,2,7 (Similarity, Pythagoras, Mensuration)

PART I — ALGEBRA

Chapter 1 — Linear Equations in Two Variables

1. Solve by the graphical method: $x + y = 5$, $x - y = 1$.
2. Solve by the elimination method: $3x + 2y = 12$, $5x - 2y = 4$.
3. Solve by Cramer's Rule: $2x + y = 8$, $x - y = 1$.
4. Solve by the substitution method: $x + 2y = 6$, $2x - y = 2$.
5. The sum of the ages of a father and son is 45 years. Five years ago, the father's age was 4 times the son's age. Find their present ages (form and solve the equations).

Chapter 2 — Quadratic Equations

6. Solve by factorisation: $x^2 - 7x + 12 = 0$
7. Solve using the formula: $2x^2 + 3x - 2 = 0$
8. Find the value of the discriminant for the equation $x^2 - 4x + 4 = 0$, and state the nature of its roots.
9. If one root of the quadratic equation $x^2 - 5x + k = 0$ is 2, find the value of k .
10. The product of two consecutive positive integers is 156. Form a quadratic equation and find the integers.

Chapter 3 — Arithmetic Progression

11. Find the first term and common difference of the AP: 5, 9, 13, 17, ...
12. Find the 15th term of the AP: 3, 7, 11, 15, ...
13. Find the sum of the first 20 terms of the AP: 2, 5, 8, 11, ...
14. Which term of the AP 4, 9, 14, 19, ... is 104?
15. How many terms of the AP 8, 15, 22, ... are needed to give a sum of 555?

Chapter 4 — Financial Planning (GST)

16. A shopkeeper sells an article for ₹1,500 (excluding GST) with a GST rate of 12%. Find the CGST and SGST amounts, and the total invoice value.
17. A dealer purchased goods worth ₹8,000 and sold them for ₹10,000, both within the same state, with GST at 18%. Find the CGST and SGST payable by the dealer to the government (input tax credit method).
18. A retailer buys a mobile phone for ₹12,000 (excluding GST at 18%) and sells it at a profit of ₹1,500 (excluding GST). Find the taxable value for the retailer and the total GST collected on the sale.

PART II — GEOMETRY

Chapter 1 — Similarity

19. State and write the Basic Proportionality Theorem (Thales Theorem).
20. In triangle ABC, $DE \parallel BC$, D and E lie on AB and AC respectively. If $AD = 3$ cm, $DB = 6$ cm and $AE = 4$ cm, find EC.
21. Triangle ABC $\sim$ Triangle PQR. If $AB = 5$ cm, $PQ = 8$ cm, and the area of triangle ABC is 50 cm^2 , find the area of triangle PQR.
22. In triangle PQR, PM is the median. If $PQ = 8$ cm, $PR = 10$ cm and $PM = 7$ cm, verify Apollonius' theorem (find $QM^2 +$ related terms).

Chapter 2 — Pythagoras Theorem

23. State and prove the Pythagoras Theorem (statement only; proof steps in brief).
24. The sides of a triangle are 9 cm, 12 cm and 15 cm. Verify whether the triangle is a right triangle.
25. In a right triangle, the two legs are 6 cm and 8 cm. Find the length of the hypotenuse.
26. A ladder 13 m long is placed against a wall such that its foot is 5 m from the base of the wall. Find the height at which the top of the ladder touches the wall.

Chapter 7 — Mensuration

27. Find the curved surface area and total surface area of a cylinder of radius 7 cm and height 10 cm. (Use $\pi = 22/7$)
28. Find the volume of a cone with radius 6 cm and height 14 cm. (Use $\pi = 22/7$)
29. Find the surface area of a sphere of radius 10.5 cm. (Use $\pi = 22/7$)
30. A cylindrical vessel of radius 5 cm and height 20 cm is completely filled with water. If all the water is poured into an empty conical vessel of the same radius and height, will the cone be able to hold all of it, or will it overflow? Justify by finding and comparing the volumes.

ANSWER KEY

PART I — ALGEBRA

Chapter 1 — Linear Equations in Two Variables

1. Adding the two equations: $2x = 6 \Rightarrow x = 3$; substituting: $y = 2$. Solution: $x = 3, y = 2$ (verify by plotting both lines — they intersect at (3, 2)).
2. Adding the two equations directly (y terms cancel): $3x + 2y = 12$ and $5x - 2y = 4 \Rightarrow 8x = 16 \Rightarrow x = 2$; substituting: $3(2) + 2y = 12 \Rightarrow 2y = 6 \Rightarrow y = 3$. Solution: $x = 2, y = 3$.
3. $D = (2)(-1) - (1)(1) = -3$; $D_x = (8)(-1) - (1)(1) = -9$; $D_y = (2)(1) - (8)(1) = -6$. $x = D_x/D = 3, y = D_y/D = 2$.
4. From eqn 2: $y = 2x - 2$. Substitute in eqn 1: $x + 2(2x - 2) = 6 \Rightarrow x + 4x - 4 = 6 \Rightarrow 5x = 10 \Rightarrow x = 2, y = 2$.
5. Let father's age = x , son's age = y . $x + y = 45$. Five years ago: $x - 5 = 4(y - 5) \Rightarrow x - 5 = 4y - 20 \Rightarrow x = 4y - 15$. Substituting: $4y - 15 + y = 45 \Rightarrow 5y = 60 \Rightarrow y = 12, x = 33$. Father = 33 years, Son = 12 years.

Chapter 2 — Quadratic Equations

6. $x^2 - 7x + 12 = 0 \Rightarrow (x-3)(x-4) = 0 \Rightarrow x = 3$ or $x = 4$.
7. $a=2, b=3, c=-2$. $x = [-3 \pm \sqrt{(9+16)}]/4 = [-3 \pm 5]/4 \Rightarrow x = 1/2$ or $x = -2$.
8. $D = b^2 - 4ac = 16 - 16 = 0$. Since $D = 0$, the roots are real and equal.
9. Substituting $x=2$: $4 - 10 + k = 0 \Rightarrow k = 6$.
10. Let integers be x and $x+1$: $x(x+1) = 156 \Rightarrow x^2 + x - 156 = 0 \Rightarrow (x-12)(x+13) = 0 \Rightarrow x = 12$ (rejecting negative). Integers: 12 and 13.

Chapter 3 — Arithmetic Progression

11. First term $a = 5$, common difference $d = 4$.
12. $t_{15} = a + 14d = 3 + 14(4) = 3 + 56 = 59$.
13. $S_{20} = 20/2 [2(2) + 19(3)] = 10[4 + 57] = 10 \times 61 = 610$.
14. $104 = 4 + (n-1)(5) \Rightarrow 100 = 5(n-1) \Rightarrow n-1 = 20 \Rightarrow n = 21$. It is the 21st term.
15. $555 = n/2 [2(8) + (n-1)(7)] \Rightarrow 1110 = n[16 + 7n - 7] = n(7n+9) \Rightarrow 7n^2 + 9n - 1110 = 0$. Solving: $n = 12$ (taking the positive root).

Chapter 4 — Financial Planning (GST)

16. GST = 12% of 1500 = ₹180. CGST = SGST = ₹90 each. Total invoice value = $1500 + 180 = ₹1,680$.
17. GST on purchase (input) = 18% of 8000 = ₹1,440 (CGST ₹720 + SGST ₹720). GST on sale (output) = 18% of 10000 = ₹1,800 (CGST ₹900 + SGST ₹900). Net GST payable = $1800 - 1440 = ₹360$, i.e., CGST payable = ₹180 and SGST payable = ₹180.
18. Selling price (taxable value) = $12000 + 1500 = ₹13,500$. GST collected on sale = 18% of 13500 = ₹2,430 (CGST ₹1,215 + SGST ₹1,215).

PART II — GEOMETRY

Chapter 1 — Similarity

19. Basic Proportionality Theorem: If a line parallel to one side of a triangle intersects the other two sides in distinct points, then the line divides those sides in proportion.
20. By BPT: $AD/DB = AE/EC \Rightarrow 3/6 = 4/EC \Rightarrow EC = 24/3 = 8$ cm.
21. Ratio of areas = (ratio of corresponding sides)². $\text{Area(PQR)}/\text{Area(ABC)} = (PQ/AB)^2 = (8/5)^2 = 64/25$. $\text{Area(PQR)} = 50 \times 64/25 = 128$ cm².
22. Apollonius' theorem: $PQ^2 + PR^2 = 2PM^2 + 2QM^2$. $64 + 100 = 2(49) + 2QM^2 \Rightarrow 164 = 98 + 2QM^2 \Rightarrow QM^2 = 33 \Rightarrow QM = \sqrt{33}$ cm.

Chapter 2 — Pythagoras Theorem

23. In a right triangle, the square of the hypotenuse is equal to the sum of the squares of the other two sides (proved using similar triangles formed by the altitude to the hypotenuse).

24. Check: $9^2 + 12^2 = 81 + 144 = 225 = 15^2$. Since this holds, the triangle is a right triangle (right angle opposite the 15 cm side).
25. $\text{Hypotenuse}^2 = 6^2 + 8^2 = 36 + 64 = 100 \Rightarrow \text{Hypotenuse} = 10 \text{ cm}$.
26. $\text{Height}^2 = 13^2 - 5^2 = 169 - 25 = 144 \Rightarrow \text{Height} = 12 \text{ m}$.

Chapter 7 — Mensuration

27. $\text{CSA} = 2\pi rh = 2 \times 22/7 \times 7 \times 10 = 440 \text{ cm}^2$. $\text{TSA} = 2\pi r(r+h) = 2 \times 22/7 \times 7 \times 17 = 748 \text{ cm}^2$.
28. $\text{Volume} = (1/3)\pi r^2 h = (1/3) \times 22/7 \times 36 \times 14 = 528 \text{ cm}^3$.
29. $\text{Surface area} = 4\pi r^2 = 4 \times 22/7 \times 10.5 \times 10.5 = 1,386 \text{ cm}^2$.
30. $\text{Volume of cylinder} = \pi r^2 h = 22/7 \times 25 \times 20 \approx 1,571.4 \text{ cm}^3$. $\text{Volume of cone} = (1/3)\pi r^2 h \approx 523.8 \text{ cm}^3$ (exactly one-third of the cylinder's volume, since both share the same radius and height). Since the cone can hold only about 523.8 cm^3 but there is $1,571.4 \text{ cm}^3$ of water, the cone CANNOT hold all the water — it will overflow.